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Introduction to Generalized Least Squares

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Assignments

Generalized Least Squares (GLS): Relations to OLS & WLS

Generalized least squares (GLS) is a statistical method used to estimate the parameters of a linear regression model when the errors of the model are not independent and identically distributed (i.i.d.).

GLS assumes that the error terms have a known covariance matrix that is not diagonal. By taking into account the covariance structure of the errors, GLS can provide more efficient and accurate estimates of the regression parameters than ordinary least squares (OLS), which assumes that the error terms are independent and have constant variance.

GLS involves transforming the original regression model into an equivalent form that has uncorrelated errors with constant variance. This is achieved by pre-multiplying both the dependent variable and the independent variables by a matrix called the "whitening matrix," which is the inverse of the square root of the covariance matrix of the errors.

After the transformation, the new regression model can be estimated using OLS, and the resulting estimates can be transformed back to the original scale using the inverse of the whitening matrix.

GLS is a powerful and flexible technique that can be used to handle a wide range of correlation structures in the errors, including serial correlation, heteroscedasticity, and spatial dependence. However, GLS requires knowledge of the true covariance structure of the errors, which may not be available in practice and needs to be estimated from the data.

The link below will walk you through step by step the derivations done in the lecture video. You will observe the mathematical set up of GLS, recovering the GLS estimator, proof that the GLS estimator is unbiased, and recovering the variance of the GLS estimator.

Please read the following article from [Towards Data Science](#) ↗ (Medium) by Andrew Rothman:

Generalized Least Squares (GLS): Relations to OLS & WLS ↗

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